

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

December 2012

CL421 Advanced Structural Analysis

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Give classification of bridges and explain components of bridges in detail. [07]

Q - 2 (a) Calculate design bending moment and design shear force for following data of deck slab. [14]

Clear span – 4 m

Width of footpath – 600 mm on either side

Wearing coat – 80 mm

Loading – IRC class A

OR

Q - 2 (a) Calculate design bending moment and design shear force for following data of deck slab. [14]

Clear span – 5.5 m

Width of footpath – 1 m on either side

Thickness of wearing coat – 100 mm

Loading – IRC class AA (Tracked)

Material – Concrete grade M 35 & Fe 415 grade HYSD bars

Q - 3 (a) Explain various loads to be considered for design of bridge. [07]

(b) Explain Corbon's Method for the analysis of a typical tee beam and slab deck bridge. [07]

OR

Q - 3 (a) Explain any two methods from the following for the analysis of a typical tee beam and slab deck bridge. [14]

1) Morrice Little Method

2) Grillage analysis

3) Hendry Jaegar Method

SECTION – II

- Q - 4 (a) What is non-linear analysis? Explain types of non-linearity. [07]
- Q - 5 (a) Enlist the assumption made in non-linear analysis. [02]
- (b) Enlist methods of solution of non-linear analysis and explain Newton-Raphson method in detail. [06]
- (c) A spherical dome having diameter 10 m and rise 3 m carries live load of 1.2 kN/m^2 . [06]
The dome has an opening of 1.4 m diameter at its crown. A lantern is provided at top, which causes a dead load of 20 kN acting along the circumference of the opening. Calculate Meridional thrust at various location of 10° interval from lantern to base of the dome.

OR

- Q - 5 (a) Figure 1 shows an assembly of three springs. Derive force displacement relationship [08]
for member AB and tangent stiffness matrix for member AB.

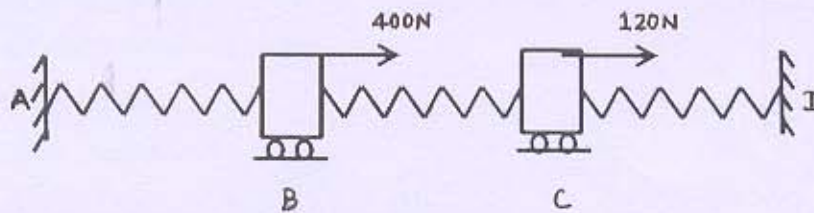


Figure: 1

For AB spring nonlinear F-D relation is $F = 100e - 10e^2$ in which e is elongation in mm and F is force in newton.

- (b) Derive an expression for Meridional thrust and hoop force for a spherical dome [06]
subjected to uniformly distributed load.
- Q - 6 (a) Define Dome. Enlist uses of dome. [04]
- (b) Design a spherical dome for a temple hall of 8 m diameter. The rise of the dome is to [10]
be 1.8 m. The live load is 800 N/m^2 . The dome carries a lantern load of 10 kN concentrated at the crown. Also give the design of ring beam. Use M 20 grade concrete and Fe 415 steel.

OR

- Q - 6 (a) Classify the dome. [06]
Explain with neat sketch Meridional thrust and hoop force.
- (b) Design a conical roof of diameter 16 m and the rise 4 m. Take the live load as 1200 N/m^2 . [08]
Use M 20 concrete and Fe 415 steel. Also give the design of ring beam.
